

QUANTITATIVE HACKATHON

GSIH 2026

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Three Quantitative Problems — Solutions & Key Insights

01

Regime Navigator

Adaptive portfolio construction across 100 stocks

EQUITIES · ALLOCATION

02

Proxy Models for Bond ETFs

Replication, risk estimation & hedging of hidden funds

FIXED INCOME · RISK

03

Pricing a Bermudan Swaption

Rates exotic priced & risk-managed from first principles

DERIVATIVES · PRICING

Regime Navigator

Allocate across **100 stocks**, rebalanced weekly — judged on **consistent, risk-adjusted growth** through market conditions we cannot see in advance, not on raw return.

PROBLEM 01
EQUITIES · ADAPTIVE ALLOCATION

1 · READ

The regime

trend, breadth & volatility



2 · RANK

Every stock

momentum, income, quality



3 · SIZE

By conviction

inverse-volatility, strict caps

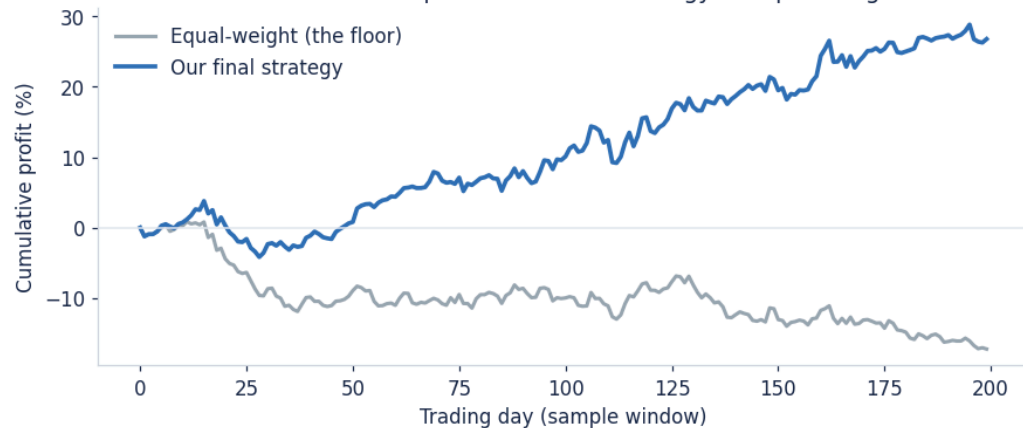


4 · PROTECT

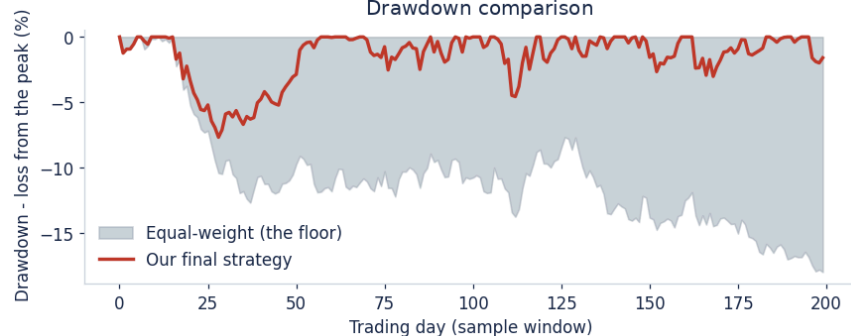
In stress

defensive tilt & hedge

Profit on the sample window: our strategy vs equal-weight



Drawdown comparison



OUR SOLUTION

- Read the market **regime**, then switch playbook with discipline — confirm before changing, so the book is never whipsawed
- Rank every stock on **momentum, dividend income, low volatility and quality**
- Size by conviction and inverse-volatility, under strict position and sector limits
- In downturns: tilt defensive, add a short hedge, and de-risk in genuine crashes

WHAT THE ANALYSIS SHOWED

- Momentum drives the returns; **income and low-volatility tilts do most to limit the losses**
- Across sample windows: roughly **double the return** of equal-weight, with a smaller worst loss
- Robust by design — fixed universe, no look-ahead, validated against extreme conditions

LIMITATIONS & WHAT'S NEXT

- Hardest case is a **correlated crash** where every stock falls together — limited room for a long book
- Add cross-asset and macro hedges, and faster regime detection
- Tuned on a limited history; more data would sharpen the switching rules

Proxy Models for Bond ETFs

Five bond funds whose holdings are **not disclosed** — rebuild each from liquid, tradeable instruments to track its value, measure its risk, and hedge a portfolio at the **lowest cost**.

PROBLEM 02
FIXED INCOME · RISK & HEDGING

1 · REPLICATE

Track value (NAV)

returns from proxy instruments



2 · MEASURE

Rate & credit risk

a sensitivity profile per fund

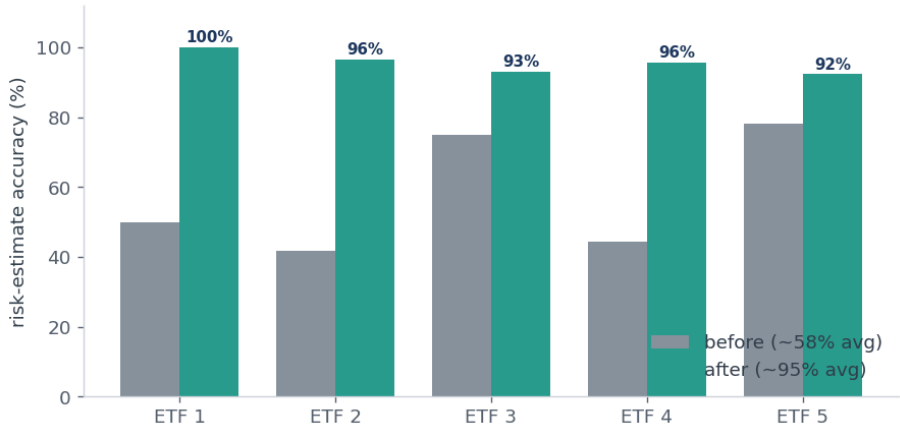


3 · HEDGE

At lowest cost

variance-minimising basket

How accurately our proxy model captures each fund's true risk



Risk accuracy. How closely our model matches each fund's true risk — accuracy roughly doubled, from ~58% to ~95% on average.

OUR SOLUTION

- Track value with a stable model of each fund's returns from liquid proxies (~99.9% accurate)
- Measure risk with a **structural rate-and-credit profile** matched to each fund's true sensitivities
- Hedge with a variance-minimising basket that trims costly, redundant legs

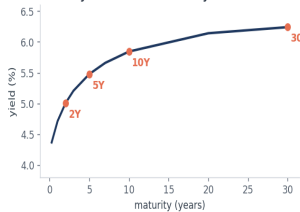
WHAT THE ANALYSIS SHOWED

- Identified each fund's character from its behaviour — from a T-bill fund to rates-plus-credit funds
- Risk-estimate accuracy **roughly doubled** (~58% → ~95%)
- Cost-aware selection removes most legs and ~half the cost with no loss of hedge quality

LIMITATIONS & WHAT'S NEXT

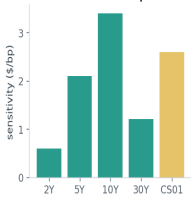
- Credit funds genuinely need ETF legs — the cash-vs-derivative basis can't be fully replicated otherwise
- Closely-offsetting (spread) positions leave an irreducible noise floor in hedging
- Extend with a wider instrument set and dynamic re-hedging

The yield curve & the 4 key-rate buckets

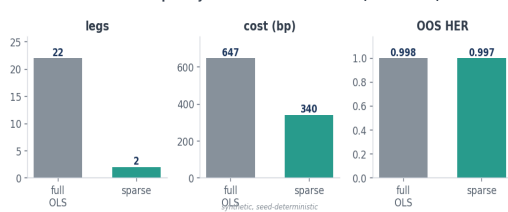


What 'risk' means. A fund's risk as a profile: sensitivity to 2/5/10/30-year rates, plus credit (CS01).

A fund's risk is a profile



Cost-aware sparsity is an anti-overfit device (5-ETF book)



Cost-aware hedge. Far fewer legs and about half the cost — with virtually identical hedge quality.

Pricing a Bermudan Swaption

An interest-rate option granting the right to enter a **pay-fixed swap on one of several future dates** — economically, a flexible position on rising rates.
Priced and risk-managed from first principles.

PROBLEM 03
RATES EXOTICS · HULL-WHITE

1 · CURVE

Build the curve

discount factors & forwards



2 · CALIBRATE

Fit the model

to the volatility surface



3 · PRICE

Early exercise

trinomial lattice

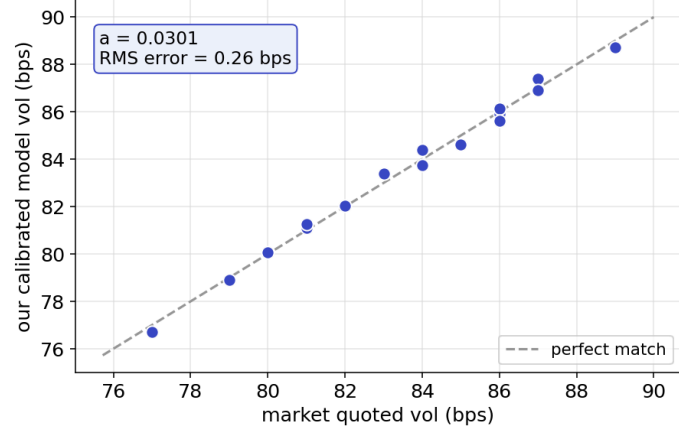


4 · RISK

Sensitivities

delta & vega

Task 2 — calibration fit: 6 dials reproduce 16 quotes



Calibration quality. Six parameters reproduce all sixteen market quotes — points sit on the perfect-match line (≈ 0.3 bp error).

OUR SOLUTION

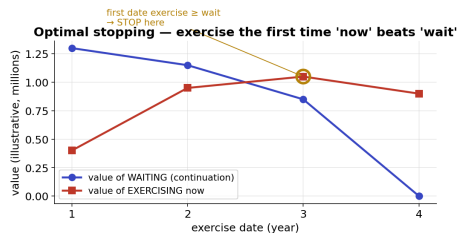
- Build the yield curve in **closed form** — arbitrage-free by construction
- Calibrate a one-factor **Hull-White** model to the full volatility surface — fast and exact
- Price the **early-exercise** feature on a trinomial lattice fitted to the curve
- Compute delta and vega by re-pricing under shifted markets

WHAT THE ANALYSIS SHOWED

- Six parameters reproduce **all sixteen market quotes to under one basis point**
- Lattice price cross-checks with a closed-form result and an independent Monte-Carlo
- Every stress scenario moves in the **economically correct direction**

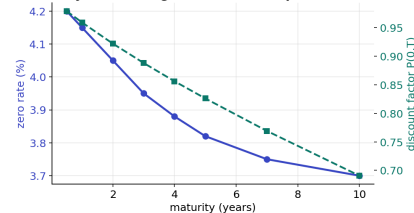
LIMITATIONS & WHAT'S NEXT

- A **flat volatility surface** makes the model's mean-reversion hard to pin down
- One factor captures the level of rates, not their changing shape
- Extend to a two-factor (G2++) or PDE engine for richer dynamics



The early-exercise decision. Wait while continuation is worth more; act the first time exercising now beats waiting.

Task 1 — the yield curve: given rates → computed discount factors



The foundation. Given rates become the discount factors that everything is valued on.